

Introduction to DDoS

Module 46

Duration: 21:49

Video: <https://www.youtube.com/watch?v=jEYv9HSp8IU>

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0:00	[Music]
0:16	dear viewers today we are going to discuss one serious kind of Cyber attack known as distributed denial of service
0:23	attack we will try to explore about introduction of Dos attacks motivation behind these attacks
0:30	what are the consequences of these attacks vulnerabilities in the basic protocols that can be exploited by the attackers to launch these attacks and in
0:39	the last part of the lecture we will discuss about the classifications of these attacks there has been a tremendous
0:47	growth of Internet especially in the last decade in terms of its usage that has led Our Lives to become increasingly
0:54	dependent on the internet the internet has become essential for information flow business Commerce communication
1:02	education entertainment and research activities initially the major goal of the internet development was to enhance
1:10	the unobstructed free flow of data and scalability without keeping mind the security issues which now appears to be
1:18	precious its architectural design Was Made Simple for ease of usage but subsequently it fascinated the attackers
1:26	who exploit the vulnerabilities and launch a variety of attacks for plethora of reasons these threats can Target
1:33	basic CIA model of security which deal with confidentiality of data Integrity of data and availability of services
1:42	availability of services is primarily targeted by launching a denial of service attacks which is characterized by an explicit attempt by attackers to
1:51	prevent legitimate users of service from using that service denial of service attacks has undoubtedly take over taken
2:00	all concerns and has emerged as arguably the most severe form of attacks in the last decade the major goal of Dos

2:08 attacks is to disrupt or deny the services for numerous reasons think out a situation you are a reputed online

2:15 shopping company and suddenly your web service becomes unavailable to the customers when you deeply investigate

2:22 you realizes that a flood of packets is surging into your network you have just become one of the hundreds or thousands

2:30 of victims of denial of service attacks a pervasive and growing threat to the internet traditionally the security

2:38 Community has focused its attention on unauthorized disclosure or modification of information and perhaps theft of

2:46 services denial of service was largely ignored as being unlikely to occur because the characters would not gain

2:53 anything from such an attack clearly this is not the case today denial of services can be devast deating attack it

3:01 3 minutes, 1 second can put Merchants out of business and can cause major and very visible disruption to our cyber world it can be

3:08 used against specific companies for which the attacker has a grudge or has been paid to attack or it can be used by

3:15 terrorists to cause major disruption to critical intermediate and victim

3:23 infrastructure this type of attack may include attempts to flood a network disrupt connections between two machines

3:30 disrupt service to a specific system or person thereby preventing legitimate Network traffic to assess Network server

3:38 service or website thus a denial of service attack aims to deny assess by legitimate users to Shar services or

3:47 resources such an attack can be configured in many ways like targeting operating system or Network

3:53 Services the first way targets to crash a system by sending one or more carefully crafted packets that exploit

4:01 4 minutes, 1 second software vulnerabilities On Target server for example the Ping of death attack that sends large International

4:08 control message protocol ping packets that is fragmented into multiple datagrams to Target system which can

4:15 cause certain operating systems to crash freeze or reboot due to buffer overflow the second way is known as

4:24 flooding type of Dos attacks that employs massive volumes of useless traffic to occupy all the resources that

4:31 could hamper service to legitimate traffic while it is possible to prevent the first form of attack by patching non

4:39 vulnerabilities however the second form of attack cannot be easily prevented the
targets can be attacked simply because

4:46 they are connected to the public internet next comes the bigger version of denial of
service attacks called

4:53 distributed denial of service attack when the traffic of Doss attacks comes from
multiple sources it is is called

5:00 distributed denial of service attack by using multiple sources the power of a Dos
tax is Amplified and the problem of

5:07 defense is made much more complicated both dos and dos are a huge threat to the
operation of internet sites but the DS problem is more complex

5:17 and harder to solve because of the following reasons first it uses a very large
number of machines this yields a

5:24 powerful weapon any Target regardless of how well provisioned it is can be taken
offline Gathering and engaging a large

5:32 army of machines has become terbly simple because many automated tools for DS
attacks can be found on hacker web

5:40 pages and in chat rooms such tools do not require sophistications to be used and
can inflict very effective damage a

5:49 large number of machines gives another advantage to an attacker even if the target
were able to identify attacking

5:56 machines and there are effective ways of fighting this information what action can
be taken against a network of 1,000

6:05 host the second characteristics of some of the Dos attacks that increase their
complexity is the use of seemingly

6:12 legitimate traffic resources are consumed by large number of legitimate looking
messages when comparing the attack messages with legitimate one

6:21 there are frequently no feature to distinguish them since the attack misus a
legitimate activity it is extremely

6:28 hard to respond to the attack without also disturbing this legitimate activity look at
the figure which shows the

6:35 complete modity of the DS attack the operating system and the network Protocols
are developed without applying security engineering which results in

6:43 providing hackers a lot of insecure machines on the internet these insecure
unpatched machines are used by Doos

6:51 attackers as the Army to launch attack an attacker or a hacker gradually implants
attack programs on these

6:58 insecure machines depending upon sophistications and logic of implanted
programs these compromised machines are called Master's

7:07 handlers or zombies and collectively called Bots and attack network is called botn
net in the hackers

7:14 Community hackers send control instructions to the Masters which in turns
communicate it to zombies for launching

7:22 attacks the zombie machines under the control of Masters and handlers as shown
in the figure transmit attack attack

7:29 packets which converge at victim or its Network to exhaust either its
communication or computational

7:38 resources numerous incidents of Dos attacks are common these days from a normal
client users to business communities isps and also the world

7:47 famous companies such as Yahoo Twitter Facebook eBay Google are the victim of
desert attacks a few recent and famous D

7:56 do incidents are shown in the table the table describes about the incidents in
chronological order right from 2000 to

8:03 2014 along with the description of attacks it is curled from the table that dos
attacks is a sensitive issue

8:11 prevailing among the cyber community and also it has been analyzed from the
recent security reports that these attacks have gained a lot of strength in

8:20 recent years that is from 2014 onwards as compared to earlier before year 2005

8:27 the attack size was less than 10 gbps and in 2010 it is reached a mark of 100 gbps
furthermore the attack size was

8:35 observed up to 400 gbps in 2014 which shows the severity of problem over the
internet now the question comes in our

8:44 mind that why a tackle launch such kinds of attacks what are their motives so in
this coming discussion we will be exploring some of the key factors that

8:53 contributes towards launching of these hostile activities on the internet first point
to be discussed about his

9:00 recognition some of the early dots attacks were largely proofs of Concepts or
simple pranks played by hackers the

9:08 ultimate goal was to prove that something could be done such as taking a popular
website offline such a major achievements brings an attacko

9:16 recognition in the underground cyber community next is ideological activism
another frequent motive of Dos attacks

9:24 is self-described as being political individuals or groups who disagree with views or
actions of certain organization

9:31 an online media site a corporation or a government have been known to launch dos
attacks against computers and networks owned by this

9:40 organization financial gain can be another factor which motivate the attackers
through the monetary gain to launch several kind of Dos attacks after

9:49 launching these attacks they may also demand for ransome amount of money to
prevent further attacks the attackers

involved these types of are most experienced next is vandalism the motive of this category of attacker is to destroy the available resources as they don't believe in anything and are senseless according to the recent reports the attackers related to this type of attacks have been increased recently lastly revenge or competitive spirit in the business attackers May launch the Dos attacks to take the Revenge the attackers may be employees of certain organizations that were Dishonored in some dispute however there is a major lack of data on preparators and motives of Dos attacks the vast majority of attacks are not reported in further discussion we will be exploring what are the consequences of these attacks the consequences of do attacks may be very severe it depends on the kind of attack launched by the attackers the organization and the target of the attackers the long duration dos attacks are more rigorous and these must be mitigated quickly after detection DS attacks may lead to the following consequences huge revenue losses reputation of organization may go down loss of shareholders confidence customers dissatisfaction loss of important data due to system crashes extra operational cost hence consequences of these attacks are so severe that it may lead to social financial and legal implication among the cyber community let's now discussed about the vulnerabilities present in our basic protocols and architecture that lead to perform this type of sophisticated attack firstly the stateless nature of the internet it means that intermediate routers do not maintain any straight information about the forwarded packets because of this accountability computer forensics are difficult to conduct that theft that Ur unidentified secondly there is a lack of authenticity over the internet without authentication malicious internet users can claim the identities of other users without being easily detected or located the most common example of this type of attack is IP spoofing where attacker can easily spoof The Source IP address of the another machine thirdly the deterministic nature of the Internet Protocol this is not a design flaw but is often necessary to proper operation of Internet protocols that is the TCP connection establishment procedure and TCP connection control mechanism the three-way handshake TCP mode mechanism which is predictable and

12:36 can be easily exploited to launch flooding type of attack known as Sin flooding attack last but not the least is the

12:44 mimicking legitimate user Behavior attackers can easily mimic the behavior of legitimate users which leads

12:52 them to perform a kind of attack called application layer dos attacks in the last last part of the

12:59 lecture we will be discussing about the classification of D attacks D do attacks may occur at every

13:07 layer of OSI model therefore these can be classified as follows as shown in the

13:14 figure physical layer attack volumetric attacks application layer attacks so first comes the physical

13:22 layer attacks also known as layer one attack these types of attacks are concerned with physical security these

13:30 attacks includes the destruction obstruction malfunctioning or manipulation of physical assets at the

13:38 physical level the communication relies on nodes host switches hubs routers and server and the medium of transmission

13:46 that can be wired or Wireless if the devices through which the services are granted gets down there will be a disruption of

13:54 services this target can be easily achieved if the signals are interrupted between communicating devices to prevent

14:02 them from reaching to their destinations so the attackers May attain their targets of denying the services by

14:09 compromising the physical security of the assets the various kinds of physical aay attacks destruction of

14:17 Assets in this kind of attack the attacker may damage the physical assets or may cut down the backbone wires of

14:24 local area network so that Services can be denied by this the availability of these assets could be reduced and there may

14:32 become unresponsive resulting in denying of services to prevent such kind of an attack physical security of assets is

14:40 required next is jamming the jamming attack is executed primarily in the wireless sensory

14:47 networks jamers can be used by attackers to hamper the functioning of the wireless networks by generating noise

14:54 signals close to the network by doing this the throughput of the network may fall drastically in Jing attacks packets

15:02 may be corrupted by generating interference between the communicating nodes next is tempering the attackers

15:11 May temper with the functioning of nodes by erasing all programs or by modifying the codes for assessing the information

15:18 of high layers cryptographic keys and other sensitive information may be extracted by coming the nodes and

15:26 keeping the packages temper approved the tempering attack can be prevented next category is volumetric

15:36 attack also known as layer two to four attacks in this attack congestion is caused by consuming the bandwidth of the

15:44 network through flooding attacks like mac flooding ICM flooding UDP flooding sin flooding

15:52 Etc therefore these attacks cause high traffic to the network a large number of DS attacks variants fall under this

15:59 category of volumetric attacks the basic protocol of this attack category is to congest the

16:06 network or Target machine with large amount of packets that is flooding the target this results in total consumption of

16:15 resources by the attacker moreover the targeted machine may become unresponsive and possibly it may crash

16:23 completely the attacker transmit such kind of amount of data toward the target Target machine through well distributed

16:31 and Scattered Bots that are remotely controlled by the bot Masters the key features of these attacks is the most of the IP packets

16:39 used have spoofed Source IP addresses the spoofing of addresses in the IP packets protect the actual attackers

16:46 from being traced back the different kinds of volumetric attacks are Mac flooding icmp flooding UDP

16:55 flooding the attackers clone the Mac addresses and send a huge number of frames of data towards the switches

17:02 besides the non-existing Mac addresses are used by The Tackle to alter the ARP

17:09 caches switches have a certain memory that is used to handle the request and to reserve the resources for

17:16 them when huge volume of forged requests are received by the switches it starts allocating the resources for the cloned

17:25 Mac addresses and start denying the request to the legit M ones likewise it may also affect the routing tables of

17:34 the routers at the network to drop the entries in the table which make the routing Services unavailable next comes the icap

17:43 flooding icmp flood is the error control protocol in the network layer query messages are used to diagnose the

17:51 network problem through icmp requests that are encapsulated into IP packet
attackers send numerous ICM MP packets

17:59 to the Target server with huge range of spoofed source IP addresses the target
servers in turn replies to these queries

18:08 which results in consumption of bandwidth on the network consequently the
network gets congested and becomes

18:16 unresponsive Additionally the fragmented icmp packets are also used by the
attackers with very high sending rate

18:24 such that server gets overloaded by This Server key keeps resembling to fake data
packets and

18:31 legitimate users do not get the services Smurf attacks and the Ping of Death attacks
are the common example which make use of

18:39 icmp protocol next is the TCP flooding the attackers exploit the vulnerability of
three-way handshake

18:48 mechanism in the TCP protocol of the transport layer to launch such kinds of attack
syn flooding attack syn flooding

18:56 attack fragmented acknowledgement packets are examples of TCP flood attack in
syn flooding attack the attackers

19:04 send large number of send requests to the Target server without spoof address to
the establish the

19:11 connection upon receiving the S request the server allocate the resources to each
request and sends response with S A

19:19 packets as responses of the server will not entertained due to the spoofed
addresses and TCP 3way handshake

19:26 connections remains in half open State till time out a very large number of such
requests

19:34 are sent by the attackers to make the target machine unavailable the scenario of
syn while flooding attack is shown in the

19:43 figure next is the UDP flood unlike TCP UDP is connectionless

19:50 Protocols of Transport layer UDP packets are used by attackers to exhaust the
bandwidth of network by launching the

19:59 attack from distributed side with the help of bots attack converges high traffic
towards the victim the DoS and the Denial of Services

20:07 of UDP are generally exploited to launch such kind of attack other kinds of attacks
which can be launched through UDP are DNS flood

20:16 UDP fragmentation vllp flood Etc final classification is based on the

20:24 application layer attacks which are also known as layer five to layer seven attacks
in contrast to volumetric attacks these

20:32 attacks generate low traffic at the network but send the large request to the web
server hence these attacks can

20:40 critically overwhelm the resources of the web server these attacks include HTTP
get
20:46 and post request flooding Etc various Protocols of the application like HTTP
20:53 https and DNS Etc are being exploited by the attacker At Last I would like to
conclude that
21:01 21 minutes, 1 second dos attacks is a devastating attacks among the cyber
community though there are many commercial Solutions present in
21:08 the market but the ideal solution to this problem is eluded so far in the next lecture
we will discuss
21:16 about the different approaches which by which we can tackle about the dasas
problem thank you
21:43 [Music]